

INDIAN STATISTICAL INSTITUTE

Mid Semester : Second Semester 2024-25

Course : M.Math First Year

Subject : Functional Analysis

Instructor : Partha Sarathi Chakraborty

Date : 1st March 2025

Marks :30

Duration: 120 minutes

Answer as much as you can. Maximum you can score is 30. Clearly state any results you will be using.

1. Let $\Omega \subseteq \mathbb{R}^d$ be an open set and $\Omega = \bigcup_{n=1}^{\infty} K_n$ be an exhaustion by compact sets.

(a) Fix a sequence $\{c_n\}_{n \in \mathbb{N}}$ of positive real numbers. Define $\rho : C_c^\infty(\Omega) \rightarrow \mathbb{R}$ by

$$\rho : C_c^\infty(\Omega) \ni f \mapsto \sum_{n=1}^{\infty} \sup_{x \in K_n \setminus K_{n-1}} \left| \frac{f(x)}{c_n} \right|,$$

where $K_0 = \emptyset$. It is easy to see and you need not waste time in showing ρ is a seminorm. Show that ρ is continuous when $C_c^\infty(\Omega)$ is endowed with the strict LF topology. [5]

(b) Show that a sequence $\{f_m\} \subseteq \mathcal{D}(\Omega)$ converges to zero only if (i) there exists n such that $\text{supp}(f_m) \subseteq K_n, \forall m$; (ii) for each multiindex $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}_0^d$

$$\lim_{m \rightarrow \infty} \sup_{x \in K_n} \left| \frac{\partial^{\alpha_1 + \dots + \alpha_d} f_m(x)}{\partial x_1^{\alpha_1} \dots \partial x_d^{\alpha_d}} \right| = 0. \quad [5]$$

2. Let E, F be finite dimensional normed linear spaces of dimension $d > 1$. Prove or disprove that $\sup\{\|T\|\|T^{-1}\| : T \in \mathcal{L}(E, F) \text{ is invertible}\} < \infty$. [5]

3. Let E, F be closed subspaces of a Banach space G . Show that $E + F$ is closed provided F is finite dimensional. (Hint: Consider the quotient map $q : G \rightarrow G/E$.) [5]

4. Let E, F be Banach spaces and $T, T_n \in \mathcal{L}(E, F), n \geq 1$ be such that $\|T_n x - Tx\| \rightarrow 0$ for all $x \in E$. Show that $\|T_n x_n - Tx\| \rightarrow 0$ whenever $\|x_n - x\| \rightarrow 0$. [5]

5. Let $\{\phi_n\}_{n \in \mathbb{N}}$ be an orthonormal sequence in a Hilbert space $\mathcal{H}$. Show that $\forall \psi \in \mathcal{H}, \langle \psi, \phi_n \rangle \rightarrow 0$ as $n \rightarrow \infty$. [5]

6. Let $S = \{\phi_n : n \in \mathbb{N}\} \subseteq L^2([0, 1], d\lambda)$ be an orthonormal set. Consider the following statements.

(a) S is an orthonormal basis.

(b) For all $x \in [0, 1]$, $\sum_{n=1}^{\infty} |\int_0^x \phi_n(t) dt|^2 = x$.

(c) $\sum_{n=1}^{\infty} \int_0^1 |\int_0^x \phi_n(t) dt|^2 dx = \frac{1}{2}$.

Show that (a) implies (b) and (c) implies (a).

[5+5=10]

☺ Good Luck ☺

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